

Foundations of Computing I

Summary

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UZH FS2026

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1 Propositional Logic

Basic Logical Equivalences

Commutative Law

- $p \wedge q \equiv q \wedge p$
- $p \vee q \equiv q \vee p$

Associative Law

- $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
- $(p \vee q) \vee r \equiv p \vee (q \vee r)$

Distributive Law

- $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
- $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$

Identity Law

- $p \wedge \mathbf{t} \equiv p$
- $p \vee \mathbf{c} \equiv p$

Negation Law

- $p \vee \sim p \equiv \mathbf{t}$
- $p \wedge \sim p \equiv \mathbf{c}$

Double Negation Law

- $\sim(\sim p) \equiv p$

Idempotent Law

- $p \wedge p \equiv p$
- $p \vee p \equiv p$

Universal Bound Law

- $p \vee \mathbf{t} \equiv \mathbf{t}$
- $p \wedge \mathbf{c} \equiv \mathbf{c}$

De Morgan's Law

- $\sim(p \wedge q) \equiv \sim p \vee \sim q$
- $\sim(p \vee q) \equiv \sim p \wedge \sim q$

Absorption Law

- $p \vee (p \wedge q) \equiv p$
- $p \wedge (p \vee q) \equiv p$

Negations of **t** and **c**

- $\sim \mathbf{t} \equiv \mathbf{c}$
- $\sim \mathbf{c} \equiv \mathbf{t}$

XOR

- $p \oplus q \equiv (p \vee q) \wedge \sim(p \wedge q)$
- $p \oplus q \equiv (a \wedge \sim b) \vee (\sim a \wedge b)$

Implication (If-Then)

- $p \rightarrow q \equiv \sim p \vee q$

Equivalence (Biconditional If and Only if)

- $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

2 Digital Logic Circuits

Disjunctive Normal Form (DNF)

“OR of ANDs”

From truth table to DNF

1. Identify the rows whose output is 1
2. AND together all variables in the same row (negate the variable when 0)
3. OR together the 1-rows

Conjunctive Normal Form (CNF)

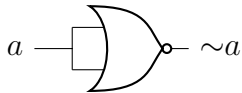
“AND of ORs”

From truth table to CNF

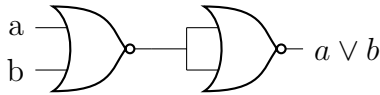
1. Identify the rows whose output is 0
2. OR together all variables in the same row (negate the variable when 1)
3. AND together the 0-rows

NOR Gate

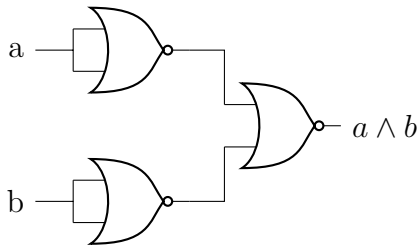
NOT Using NOR



OR Using NOR

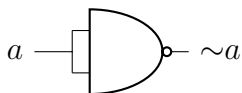


AND Using NOR

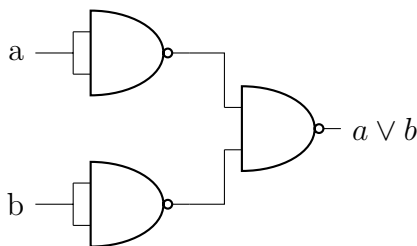


NAND Gate

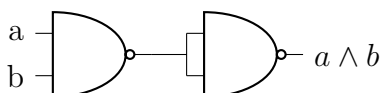
NOT Using NAND



OR Using NAND



AND Using NAND



3 Induction and Recursion

Proof by Mathematical Induction

To prove that a property $P(n)$ is **true** for every integer $n \geq b$:

1. **Base Case:** Show that $P(b)$ is true

2. **Inductive Hypothesis:** Assume $P(k)$ true for any $k \geq b$
3. **Inductive Step:** Show that $P(k) \rightarrow P(k+1)$ is **true**
4. **Conclusion:** If $P(b)$ is true and $P(k) \rightarrow P(k+1)$ is true, conclude that $P(n)$ is true for all $n \geq b$

Proof by Strong Mathematical Induction

- Prove recursive problems with several base cases (e.g., recurrence relations with order larger than 1)
- Makes a stronger hypothesis than the principle of Mathematical Induction:
 - The **base step** may contain proofs for **several initial values**, rather than just the base case
 - In the inductive step, the trust of $P(k)$ is assumed not just for an arbitrary $k \geq b$ but for **all values through k**

Let b, b be two integers with $a \leq b$. To prove that a property $P(n)$ is **true** for every integer $n \geq a$:

1. **Base Case:** Show that $P(n)$ is true for all base cases from a to b , that is: $P(a) \wedge P(a+1) \wedge \dots \wedge P(b)$
2. **Inductive Hypothesis:** For any $k \geq b$, assume $P(i)$ is **true** for $i = a, \dots, b, \dots, k$
3. **Show** that $P(k+1)$ is **true**
4. **Conclusion:** If 1 and 2 are verified, conclude that $P(n)$ is true for all $n \geq a$

Method of the Loop Invariant to Prove Algorithm Correctness

For every $n \geq 0$, let $P(n)$ be the loop invariant for a loop. **If the following four properties are true, then the loop is correct** with respect to its pre- and post-conditions:

1. **Basic property:** the loop invariant is true before the loop starts, i.e., $P(0)$ is true before the loop starts. ($P(0)$ is $P(n)$ when $n = 0$). $\rightarrow$ “The loop invariant is true before the loop starts (i.e., base case is true)”
2. **Inductive property:** for any integer $k \geq 0$, if **both the guard G and the loop invariant $P(k)$** are true before an iteration of the loop, then $P(k + 1)$ is true after iteration. $\rightarrow$ “It remains true after every iteration (i.e., $P(k) \rightarrow P(k + 1)$)”
3. **Eventual Falsity of the Guard:** after a finite number of iterations of the loop, the guard G becomes false. $\rightarrow$ “The loop eventually stops (i.e., that the loop condition eventually turns false)”
4. **Correctness of the post-condition:** if N is the number of iterations after which G is false and $P(n)$ is true, then the post-condition is satisfied. $\rightarrow$ “At the point when the loop ends, you need to verify that the **loop invariant** is true”

4 Functions and Convolution

Laws of Exponents

If b and c are any positive real numbers and u and v are any real numbers, the following laws of exponents hold true:

- $b^u b^v = b^{u+v}$
- $(b^u)^v = b^{uv}$
- $\frac{b^u}{b^v} = b^{u-v}$
- $(bc)^u = b^u c^u$

Laws of Logarithms

For any positive real numbers b , c and x with $b \neq 1$ and $c \neq 1$:

- $\log_b(xy) = \log_b x + \log_b y$
- $\log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y$

- $\log_b(x^a) = a \log_b x$
- $\log_c x = \frac{\log_b x}{\log_b c}$

1D Convolution

$$f(n), g(n), y(n): \mathbb{Z} \rightarrow \mathbb{R} \quad \forall n \in \mathbb{Z}$$

1D convolution between f and g , denoted with $y(n) = (f * g)(n) = \sum_{k=-\infty}^{+\infty} f(k)g(n - k)$

- f : input function
- g : kernel or filter

To convolve two functions f and g : $f * g$

1. Flip g horizontally ($-k$)
2. Slide it over f by adjusting n and compute the sum of the products, starting at the first overlap
3. Continue until the last overlap
4. Calculate $y(n)$ for using all n -values used to slide g over f

Note: Since the operation is commutative, sometimes it might be more convenient to compute $g * f$

Convolution Table Approach

1. Create vector representations of f and g
2. Identify *min index* and *max index* $\rightarrow$ min and max non-zero values on the x -axis for f and g
3. Create a table with f as columns from *min index* to *max index* and g as rows from *min index* to *max index*
4. Multiply the values
5. Create $y(n)$ by summing the diagonals
6. $n = 0$ at the diagonal where *row index* + *column index* = 0 (and equivalent for other n -values)

Properties of Convolution

- **Commutative Law:** $f * g = g * f$
- **Associative Law:**
 $(f * g) * h = f * (g * h)$

- **Associative Law with Scalar Multiplication:**

$$(\lambda f) * g = f * (\lambda g) = \lambda(f * g)$$

- **Distributive Law:**

$$f * (g + h) = (f * g) + (f * h)$$

- **Convolution with a Dirac:** $f * \delta = f$

2. Slide the filter over the image, starting in the top right
3. Compute the sum of products (replace each pixel with a weighted sum of its neighbors)
4. Continue towards the right until the end of the row
5. Move to the next row, starting from the left again

Why We Flip the Function in the Definition of Convolution

$$(f * g)(n) = (g * f)(n) \Rightarrow \sum_{k=-\infty}^{+\infty} f(k)g(n-k) = \sum_{k=-\infty}^{+\infty} g(k)f(n-k)$$

- Without flipping, convolution would not be commutative

Note: If we don't want to change the intensities, the sum of all elements in the filter should be 1.

Dirac Function

Also called *impulse function*. The result of the convolution of a function $f(n)$ with a Dirac is the function itself: $(f * \delta)(n) = f(n)$

$$\delta(n) = \begin{cases} 1 & n = 0 \\ 0 & \text{elsewhere} \end{cases}$$

or

$$\delta(n-i) = \begin{cases} 1 & n = i \\ 0 & \text{elsewhere} \end{cases}$$

Note that $f(n) * \delta(n-i)$ shifts $f(n)$ by i to the right.

2D Convolution

Let F, G, Y be **functions** defined on the infinite set $\mathbb{Z}^2$: $F(i, j), G(i, j), Y(i, j)$:
 $\mathbb{Z}^2 \rightarrow \mathbb{R} \quad \forall(i, j) \in \mathbb{Z}^2$

We define the 2D convolution between f and g : $Y(i, j) = (F * G)(i, j) = \sum_{u=-\infty}^{+\infty} \sum_{v=-\infty}^{+\infty} F(u, v)G(i-u, j-v)$

- f : input image
- g : kernel or filter

Box Filters

1. Flip the filter in both dimensions (= 180 deg turn)

Example Low-Pass Filter

$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$

Example High-Pass Filter

0	1	0
1	-4	1
0	1	0

Example Horizontal (x) Axis Filter

-1	0	1
-1	0	1
-1	0	1

Example Vertical (y) Axis Filter

-1	-1	-1
0	0	0
1	1	1

Boundary Issues

- **Zero-padding** of the input image boundaries to achieve the **same size** of the input and output image.

- **No zero-padding** of the input image results in a **smaller size** of the output image.

Filters of Even Side Length

- If our filter has even side length, we need to determine the center out of 4 options. By convention, we chose the point at the bottom right (option 4)
- If we add zero-padding, this needs to be adapted according to where the center is chosen

	1	2	
	3	4	